

Joint Power and Trajectory Optimization in UAV Anti-jamming Communication Networks

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Abstract—This paper mainly investigates the unmanned aerial vehicle (UAV) communication networks under the threat of a static malicious jammer. While taking the flying process of the user (UAV transmitter-receiver pair) into consideration, we propose a joint power and trajectory optimization method. Moreover, a Stackelberg framework is formulated to solve the proposed optimization problem. In addition, a joint power and trajectory optimization algorithm (JPTOA) based on best-response (BR) is designed to obtain the user's strategy as well as Stackelberg equilibrium (SE) in each time stage. Furthermore, as an extension of single stage optimization, the I -step exploration and one-step decision (IEOD) scheme is designed to enhance the user's cumulative utility. Finally, simulation results are presented to show the performance of the proposed JPTOA scheme.

Index Terms—UAV anti-jamming communication networks, Stackelberg game, joint power and trajectory optimization.

I. INTRODUCTION

JAMMING attacks have put great threat on unmanned aerial vehicle (UAV) communication networks. In anti-jamming field, various techniques have been applied in power domain [1]–[3], frequency domain [4]–[10] and multi-domain [11], with little concern about the UAV communication networks. Facing malicious jamming, it is hard for UAVs to establish reliable communication because of the large-scale and high dynamic characteristics of UAV communication.

Considering the competitive interactions between user-side and jammer-side, game theory can well describe the players' interactions and decisions, and it has been employed to model the anti-jamming problem specifically [12]–[16]. In [17], a stochastic game was formulated to make anti-jamming defense, while the hierarchical characteristic between user and jammer has not been investigated. In [18]–[27], the Stackelberg game framework, which is suitable to model the competitive hierarchical characteristics, was proposed to investigate the anti-jamming problem. Moreover, in [28], the authors investigated the anti-jamming transmission in UAV communication networks, taking multi-user case into consideration. In [29], to improve the communication performance of VANETs, the authors used unmanned aerial vehicles (UAVs) to relay messages and studied the anti-jamming effectiveness of the proposed method. However, most existing anti-jamming studies scarcely considered the mobility in UAV communication networks. Besides, although anti-jamming power optimization has been

widely studied, none of these works considers the joint power and trajectory optimization in anti-jamming field.

In fact, comparing anti-jamming UAV communication networks with traditional anti-jamming field, we think there are two factors that should be concerned. Firstly, mobility is an important factor that should be taken into account for the UAV communication networks, which was hardly considered in existing works. Hence it is necessary for UAV to consider its flight trajectory under the jamming environment. Secondly, due to the limited energy resource for the UAV, it is also significant to project its flight trajectory for the purpose of improving energy efficiency. To solve these two challenges, in this paper, we mainly investigate the joint power and trajectory optimization problem in UAV anti-jamming communication networks using the Stackelberg framework. For the user, it aims to find the best strategy to realize reliable transmission as well as to improve energy efficiency, so the user's utility is formulated as the tradeoff among throughput, transmission cost and flying cost. While for the jammer, it has the purpose to degrade the user's transmission performance with a considerably low cost. To sum up, the main contributions of this paper are:

- A joint power and trajectory optimization anti-jamming scheme is proposed using a Stackelberg game framework.
- The Stackelberg equilibrium (SE) for single stage game optimization is derived, and an I -step exploration and one-step decision (IEOD) scheme is employed to enhance the user's cumulative utility.
- A joint power and trajectory optimization algorithm (JPTOA) based on best-response (BR) is designed. Moreover, simulation results exhibit the performance of the JPTOA scheme.

Compared to works [30], [31], this paper contains differences as follows: i) We mainly consider the joint power and trajectory optimization situation for the user instead of investigating the power strategies unilaterally. ii) To enhance the user's cumulative utility, an IEOD scheme is newly designed.

The rest of the paper is organized as follows. The system model and problem formulation are presented in Section II. In Section III, the joint power and trajectory optimization anti-jamming problem is formulated as a Stackelberg game, and the corresponding joint power and trajectory optimization

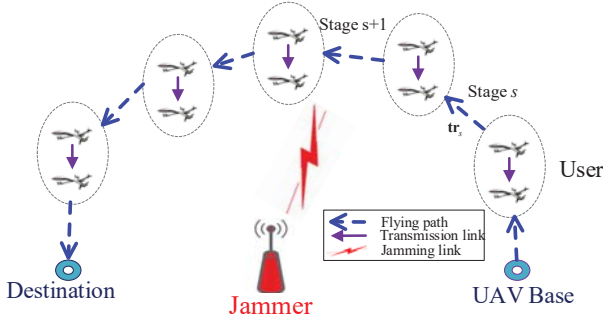


Fig. 1. Flying UAVs under the threat of the jammer.

algorithm is proposed in Section IV. Simulations results are shown and analysed in Section V. Finally, we draw the conclusion in Section VI.

II. SYSTEM MODEL AND PROBLEM FORMULATION

The system model is shown in Fig.1. It is assumed that there is one user (a two-UAV formation) and one jammer competing with each other. The user aims to communicate with the receiver while flying to its destination. Specifically, denote $\mathcal{P} = [p_1, \dots, p_N]$ as the user's power set, $\mathcal{J} = [j_1, \dots, j_N]$ as the jammer's power set, $\mathcal{S} = [1, \dots, S]$ is the flying time stage set. Motivated by [19] [20], and taking transmission cost as well as flying cost into consideration, we design the user's utility in time stage s , $s \in \mathcal{S}$ based on Signal-to-Interference-plus-Noise Ratio (SINR), showing as:

$$r_u(s) = \frac{p_s H_u(s) t_s}{N_0 + j_s H_j(s)} - c_u p_s t_s - c_f d_f(s). \quad (1)$$

In time stage s , $p_s \in \mathcal{P}$ is the user's transmission power, and $j_s \in \mathcal{J}$ is the jammer's jamming power. $H_u(s)$ denotes the channel power gain of the transmission link, while $H_j(s)$ is the channel power gain of the jamming link. The time duration t_s of each stage s is assumed to be constant, and N_0 denotes the noise power. Moreover, c_u is the user's transmission cost, and c_f is the flying cost per unit distance. Furthermore, the trajectory set in each time stage s is $\mathcal{M} = [\mathbf{tr}_1, \dots, \mathbf{tr}_M]$, which means the user has M possible trajectories in stage s . The flying trajectory distance in stage s is $d_f(s)$. Thus, the user's cumulative utility in the whole flying and transmission process is:

$$R_u = \sum_{s=1}^S r_u(s). \quad (2)$$

From the jammer's perspective, the purpose is to deteriorate the user's transmission performance once the user starts transmitting. When transmission cost is considered, the jammer's utility in time stage s is denoted as:

$$r_j(s) = -\frac{p_s H_u(s) t_s}{N_0 + j_s H_j(s)} - c_j j_s t_s, \quad (3)$$

where c_j is the jammer's transmission cost per unit power. Then the jammer's cumulative utility is denoted as:

$$R_j = \sum_{s=1}^S r_j(s). \quad (4)$$

Hence, the user's global optimization object is to maximize its cumulative utility as well as find the best power strategy and trajectory strategy combination. Denote the user's global strategy as $A_u = [p_1, \mathbf{tr}_1, p_2, \mathbf{tr}_2, \dots, p_S, \mathbf{tr}_S]$, where $\mathbf{tr}_s \in \mathcal{M}$ is the user's trajectory in stage s , and satisfies $|\mathbf{tr}_s| = d_f(s)$. Denote the jammer's strategy as $A_j = [j_1, \dots, j_S]$, then the global optimization problem is expressed as:

$$P: \begin{aligned} A_u &= \arg \max R_u(A_u, A_j), \\ A_j &= \arg \max R_j(A_u, A_j). \end{aligned} \quad (5)$$

In detail, the user's strategy combination A_u contains two decisions: Firstly, it chooses the best transmission power p_s at each time stage s ; Secondly, based on the utility function, it chooses its optimal flying strategy $\mathbf{tr}_s$ in every stage s . While for the jammer, to cope with the transmission strategy made by the user in every stage s , the jammer will employ the best jamming strategy as a response.

However, as the strategy space of the global optimization problem is too huge (about $(M \times N)^S$), it is hard to find the best solution. Thus, we turn to find the best optimization solution at each stage s , and the optimization problem for stage s is:

$$P(s): \begin{aligned} a_u(s) &= \arg \max r_u(a_u(s), a_j(s)), \\ a_j(s) &= \arg \max r_j(a_u(s), a_j(s)). \end{aligned} \quad (6)$$

where $a_u(s) = [p_s, \mathbf{tr}_s]$, $a_j(s) = j_s$ are the user's and jammer's strategy in stage s .

Motivated by [32], we use the line-of-sight (LoS) channel model to formulate for the UAV transmission link, and use the non-line-of-sight (NLoS) channel model to formulate for the jammer-UAV link, the path loss can be expressed as:

$$\begin{cases} PL_u = \left(\frac{4\pi f_c d_0}{c} \right)^2 (d_u/d_0)^2 \mu_{\text{LoS}}, \text{ LoSlink}, \\ PL_j = \left(\frac{4\pi f_c d_0}{c} \right)^2 (d_j/d_0)^2 \mu_{\text{NLoS}}, \text{ NLoSlink}, \end{cases} \quad (7)$$

where PL_u and PL_j are the path loss of UAV-UAV link and jammer-UAV link respectively, f_c is the carrier frequency, and d_0 is the reference distance. Moreover, d_u and d_j are the distance of UAV-UAV link and jammer-UAV link, c is the speed of light, μ_{LoS} and μ_{NLoS} are attenuation factors for LoS and NLoS link respectively. Hence, the channel power gain in time stage s can be denoted as:

$$\begin{cases} H_u(s) = 1/PL_u(s), \\ H_j(s) = 1/PL_j(s). \end{cases} \quad (8)$$

III. ANTI-JAMMING STACKELBERG GAME FOR JOINT POWER AND TRAJECTORY OPTIMIZATION

In this part, the joint power and trajectory optimization anti-jamming problem is formulated as a Stackelberg framework. The global game in the whole time stage S is firstly introduced, then the single stage game is proposed, and the Stackelberg Equilibrium is also defined.

In the Stackelberg game, the user is assumed to be the leader, while the jammer performs as the follower. Mathematically, denote the global game as:

$$\mathcal{G} = \{\{u, j\}, \{\mathbb{A}_u, \mathbb{A}_j\}, \{A_u, A_j\}, \{R_u, R_j\}\}, \quad (9)$$

where u and j denote the user and the jammer. $\mathbb{A}_u$ and $\mathbb{A}_j$ are the strategy sets of the user and the jammer. In addition, $A_u \in \mathbb{A}_u$ and $A_j \in \mathbb{A}_j$ are the user's and jammer's global strategy choice, R_u and R_j represent the user's and jammer's utility respectively.

Nevertheless, due to the huge strategy space of the global game $\mathcal{G}$, it is hard to find the best solution. Thus, we turn to find the best optimization solution at each single stage s . The stage game in time stage s can be expressed as:

$$G(s) = \left\{ \begin{array}{l} \{u, j\}, \{\Lambda_u(s), \Lambda_j(s)\}, \\ \{a_u(s), a_j(s)\}, \{r_u(s), r_j(s)\} \end{array} \right\}, \quad (10)$$

where $r_u(s)$ and $r_j(s)$ are the players' stage utility formulated in Eq. (1) and Eq. (3), and user's strategy set $\Lambda_u(s) = \mathcal{P} \otimes \mathcal{M}$, jammer's strategy set $\Lambda_j(s) = \mathcal{J}$. Denote $a_u^*(s) \in \Lambda_u(s)$ as the best strategy of the user in stage s , and $a_j^*(s) \in \Lambda_j(s)$ as the jammer's best response, the Stackelberg equilibrium (SE) in stage game $\mathcal{G}(s)$ is expressed as follows.

Definition 1. Motivated by [19], [20], [28], the strategy combination $a_u^*(s), a_j^*(s)$ constitutes the SE, if $a_u^*(s)$ maximizes the user's utility in stage s , and $a_j^*(s)$ is the jammer's best response strategy to $a_u^*(s)$. For any strategy combination $(a_u(s), a_j(s))$, the following conditions are satisfied:

$$\begin{aligned} r_u(a_u^*(s), a_j^*(s)) &\geq r_u(a_u(s), a_j^*(s)), \\ r_j(a_u^*(s), a_j^*(s)) &\geq r_j(a_u^*(s), a_j(s)). \end{aligned} \quad (11)$$

Stackelberg game is a kind of extension of traditional non-cooperative game, and has distinct hierarchical feature. When the user-side and the jammer-side achieve SE, none of them has the motivation to deviate from it.

Theorem 1. For every stage s , there exists SE in the anti-jamming joint power and trajectory optimization Stackelberg game.

Proof: Inspired by [30], for a given strategy $a_u^*(s)$, the jammer's game degrades into a follower sub-game $\mathcal{G}_f(s) = \{\Lambda_j(s), a_j(s), r_j(s)\}$. As we assume that there is one jammer in the system model, and the jammer's power strategy set $\Lambda_j(s) = \mathcal{J}$ is finite, then the jammer's best response strategy which maximizes its utility is the best strategy of the follower sub-game.

Denote $BR(a_u(s))$ as the jammer's best response strategy when the user's strategy $a_u(s)$ is given, then SE can be equivalently denoted as:

$$r_u(a_u^*(s), BR(a_u^*(s))) \geq r_u(a_u(s), BR(a_u(s))). \quad (12)$$

In addition, the user's strategy space is $N \times M$, which is also finite. Thus, we can find the user's best strategy $a_u^*(s)$, which is the upper bound of its utility function, shown as follows:

$$r_u(a_u^*(s), BR(a_u^*(s))) = \sup_{a_u(s)} r_u(a_u(s), BR(a_u(s))). \quad (13)$$

Thus, in our proposed single game $\mathcal{G}(s)$, there always exists SE. For better clarification, the flow of obtaining SE is shown in Fig. 2. ■

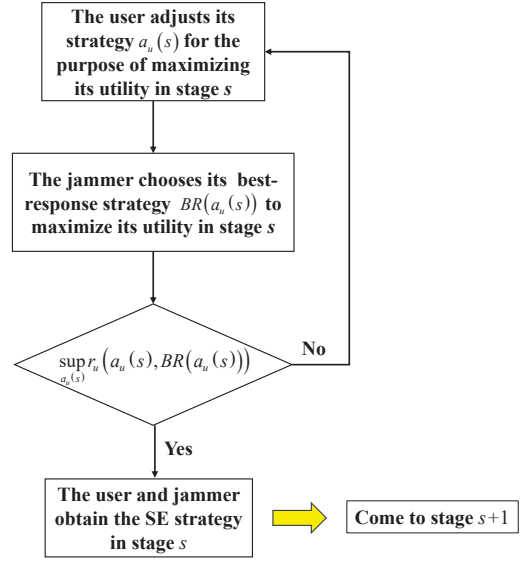


Fig. 2. The process of obtaining SE in stage s .

IV. JOINT POWER AND TRAJECTORY OPTIMIZATION ALGORITHM

A. Algorithm Description

In this section, the joint power and trajectory optimization algorithm (JPTOA) based on best-response (BR) is proposed. We assume that players are smart enough to make strategy decisions adaptively. In Algorithm 1, the single stage optimization algorithm is shown. In stage s , the user firstly chooses a strategy $a_u(s, m, n)$ from its strategy set $\Lambda_u(s)$. When the jammer senses the user's strategy, it chooses best response strategy from its strategy set $\Lambda_j(s)$, based on the following equation:

$$a_j^*(s, m, n) = \arg \max r_j(a_u(s, m, n), a_j(s, m, n)), \quad (14)$$

where $a_u(s, m, n)$ denotes that the user chooses power strategy p_n and trajectory strategy tr_m in stage s , while $a_j(s, n)$ is the corresponding jamming strategy of the jammer.

As the leader of the game, the user has first-mover advantage. Thus, the user will adopt the best strategy $a_u^*(s, m, n)$, which can maximize its utility, based on the following equation:

$$a_u^*(s, m, n) = \arg \max r_u(a_u(s, m, n), a_j^*(s, m, n)). \quad (15)$$

After the Algorithm 1 is done, the SE in stage s contains the best strategy for the user and jammer. Moreover, the user's maximal stage utilities in all S stages constitute the cumulative utility, and the trajectory strategies in all S stages constitute the whole trajectory from the UAV base to its destination.

B. I-step Exploration and One-step Decision Scheme

In this part, we design the I -step exploration and one-step decision (IEOD) scheme as the extension of single-step optimization ($I = 1$), which improves the user's cumulative

Algorithm 1: Single Stage Joint Power and Trajectory Optimization
Algorithm Based on Best-Response

Initialization: Initialize user's power set $\mathcal{P}$, jammer's power set $\mathcal{J}$, position set $\mathcal{M}$, noise power N_0 , transmission cost c_u , c_j and flying cost c_f .

Repeat Interactions:

- (1) **for** $s = 1 : S$
- (2) **for** $m = 1 : M$
- (3) **for** $n = 1 : N$
- (4) The user chooses its strategy $a_u(s, m, n)$ from its strategy set $\mathcal{M} \times \mathcal{P}$. ($a_u(s, m, n)$ denotes trajectory strategy $\mathbf{tr}_m$ and power strategy p_n)
- (5) The jammer chooses its best-response strategy $a_j^*(s, m, n)$, which maximizes its utility when sensing $a_u(s, m, n)$.
- (6) **end for**
- (7) **end for**
- (8) The user finds the SE strategy $(a_u^*(s), a_j^*(s))$ in all strategy combination $(a_u(s, m, n), a_j^*(s, m, n))$, and finally chooses strategy $a_u^*(s)$ in stage s .
- (9) **end for**

Output:

- (1) Save the SE strategy and corresponding utility in each stage.
 - (2) Calculate the user's cumulative utility in all stages.
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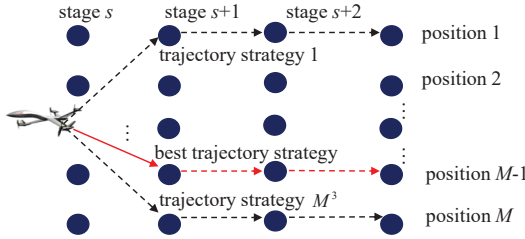


Fig. 3. I -step exploration and one-step decision scheme ($I = 3$).

utility. The user's optimization object in IEOD scheme turns to be:

$$P_{IEOD} : \max_{a_u(s)} \sum_{i=s}^{s+I-1} r_u(a_u(i), a_j(i)). \quad (16)$$

For better description, the case $I = 3$ is shown in Fig. 3, where the user tries to do a three-step exploration and one-step decision. The total strategy space increases to $(N \times M)^3$, and we can find the SE space using the JPTOA algorithm in the extended strategy space. Thus, the single stage JPTOA algorithm can be extended to IEOD scheme by increasing the strategy space to $(N \times M)^I$.

Based on the JPTOA-IEOD scheme, the user can find the best I -step exploration strategies which contains the best power strategy and best trajectory strategy from its strategy set. And then the user adopts the first step decision as its strategy in current stage s . After that, the user makes another I -step exploration in the next stage $s + 1$. Until there are less than I stages, the user chooses to do an $(I - 1)$ -step exploration optimization until it arrives at its destination.

V. SIMULATION RESULTS AND DISCUSSIONS

In this section, simulation results are presented. The power sets of the user and jammer are:

$$\begin{cases} P = \{1\text{mW}, 1.5\text{mW}, 2\text{mW}, \dots, 5.5\text{mW}\}, \\ J = \{36\text{W}, 36.5\text{W}, 37\text{W}, \dots, 40.5\text{W}\}. \end{cases} \quad (17)$$

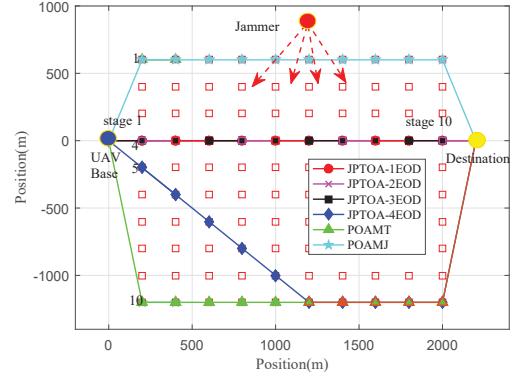


Fig. 4. The trajectory comparison of different schemes.

Assuming that the flight altitude is constant, we use a 2D scenario to simulate. The jammer is located in (1200m, 4000m) statically, whereas the user flies from (0m, 0m) to (2200m, 0m). The flying process is divided into $S = 10$ time stages, the time duration of each stage is $t_s = 5\text{s}$. For each stage, the user has $M = 10$ trajectories to choose. The parallel distance between two adjacent stages is 200m, and the distance of the transmitter-receiver pair is $d_u = 100\text{m}$. In addition, the channel parameters are referred to [32], which introduces a 2GHz carrier frequency. In detail, $d_0 = 1\text{m}$, $\mu_{\text{LoS}} = 3\text{dB}$, $\mu_{\text{NLoS}} = 23\text{dB}$, $N_0 = -200\text{dBm}$. In addition, $c_u = 10/\text{power}$, $c_j = 1/\text{power}$, and $c_f = 0.045/\text{m}$.

To make a evaluation of JPTOA algorithm, it is compared with the POAMT method and POAMJ method. In POAMT method, the power strategy is obtained based on Stackelberg framework, while the minimum trajectory strategies are employed, and in POAMJ method, the power strategy is also obtained via Stackelberg framework while the minimum jamming trajectory strategies are adopted. Fig. 4 shows the trajectory strategies of six algorithms: The red line, pink line and black line are the trajectory of 1EOD ($I = 1$), 2EOD ($I = 2$) and 3EOD ($I = 3$) scheme respectively, which depict that these three methods adopt the same trajectory strategy. The blue line shows the trajectory of 4EOD scheme, the wathet line presents the trajectory of POAMT scheme, the green line represents the trajectory of POAMJ scheme. Besides, the orange line denotes the overlapping trajectory of 4EOD scheme and POAMJ scheme.

Fig. 5 depicts the user's cumulative utility under six different schemes. In addition, we compare the JPTOA algorithm with POA scheme, showing that the JPTOA-4EOD scheme achieves the highest cumulative utility. The POAMJ scheme obtains higher cumulative utility than 1EOD, 2EOD and 3EOD schemes as these three methods find the located optimal solution due to the limited strategy space. With the increase of time stages, the advantage of 4EOD scheme begins to emerge. Hence, the proposed JPTOA scheme is an effective way to enhance the anti-jamming ability for UAVs.

VI. CONCLUSION

In this paper, we investigated the joint power and trajectory optimization problem in UAV anti-jamming communication

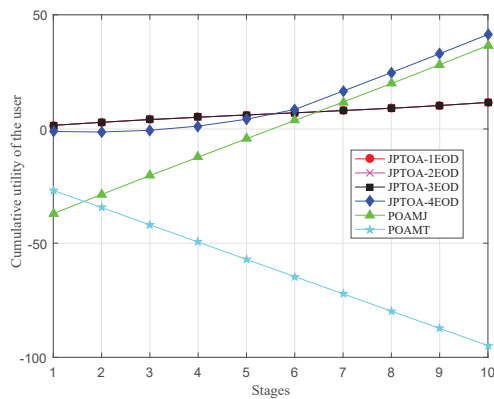


Fig. 5. Performance comparison of the user's cumulative utility for different schemes.

networks. A Stackelberg game framework was proposed, and an I -step exploration and one-step decision scheme was adopted to enhance the user's cumulative utility. Simulations were conducted for the joint power and trajectory optimization algorithm (JPTOA) based on best-response. Moreover, the trajectory strategies were presented, and the user's cumulative utilities of different schemes were compared. Finally, simulation results indicated the necessity of joint power and trajectory optimization, and the effectiveness of IEOD scheme.

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